

CRGS

Customer Recovery & Growth System

Project Presentation Report

Web Admin Portal • REX Field Sales Mobile App • Flask / Oracle API

Prepared for stakeholder presentation

July 2026

Version 1.0

Platform summary

CRGS connects sales managers (web) and field executives (mobile) through a shared Flask API on Oracle ERP data — enabling route execution, customer recovery, target tracking, visit capture, and growth campaigns.

01 CONTENTS

Table of Contents

1. Executive Summary
2. Problem Statement & Business Objectives
3. Solution Overview — CRGS & REX APP
4. System Architecture
5. Technology Stack
6. Web Admin Portal — Modules & Features
7. REX Mobile App — Modules & Features
8. Backend API & Data Layer
9. Domain Model & User Roles
10. End-to-End Operational Flow
11. UX / UI Design Principles
12. Integration Map (Web ↔ Mobile ↔ Oracle)
13. Current Maturity & Roadmap Notes
14. Demo Script for Presentation
15. Conclusion

01 EXECUTIVE SUMMARY

1. Executive Summary

The **Customer Recovery & Growth System (CRGS)** is a field-force operations platform designed for sales organizations that sell through named routes and customer accounts. It addresses three persistent operational gaps: loss of inactive (“missing”) customers, weak visibility of field execution against targets, and fragmented coordination between office managers and executives on the road.

The platform ships as two tightly coupled client applications over one shared backend:

- **CRGS Admin Web Portal** — for sales managers: user management, route assignment, multi-dimensional targets, task campaigns, dashboards, and reports.
- **REX APP (Mobile)** — for sales executives: route execution, customer triage (missing / outstanding), GPS-backed visits, product introduction, expected orders, tasks, and personal performance views.
- **Flask REST API + Oracle DB** — authenticates users, exposes ERP-backed customers and items, and persists CRGS operational tables (targets, tasks, visits, orders).

Together, these components close the loop from **plan** → **assign** → **execute** → **measure**, giving managers control from the office and executives a guided workflow in the field.

02 PROBLEM & OBJECTIVES

2. Problem Statement & Business Objectives

2.1 Challenges in traditional route sales

- Customers stop purchasing without a structured recovery process (product issues, competitor switch, credit holds).
- Managers lack a single view of executive load, route coverage, and target achievement.
- Field visits and expected orders are captured inconsistently or on paper/spreadsheets.
- Outstanding collections and follow-ups are hard to prioritize during a busy route day.
- Campaign work (new products, replacements, market research) is assigned verbally and not tracked.

2.2 Business objectives

Objective	How CRGS delivers
Recover missing customers	Missing filters (7–60 days), recovery forms, follow-up tasks
Grow revenue	Sales / product / customer targets + product introduction flows
Improve route discipline	Assigned routes, visit start/end with GPS & duration
Manager visibility	Live dashboard KPIs, task overdue counts, multi-tab reports
Field productivity	Mobile-first triage (All / Missing / Outstanding) + maps
Unify office & field	Shared Oracle data via one API for both clients

03 SOLUTION OVERVIEW

3. Solution Overview — CRGS & REX APP

Product	Audience	Primary job
CRGS Admin Portal	Sales Manager / Admin	Plan workforce: create executives, assign routes, set targets & tasks, monitor performance
REX APP	Sales Executive	Execute routes: visit customers, recover accounts, introduce products, capture orders
crgs-admin-api	Shared service	Secure REST access to Oracle RFSS/rgc ERP views and CRGS_* operational tables

3.1 Naming & branding

- **CRGS** = Customer Recovery & Growth System (portal tagline on Login & Layout).
- **REX APP** = mobile companion branded as a Field Sales Platform (pubspec / app constants).
- Shared brand color: teal #0F766E with orange accent for primary mobile CTAs.

3.2 Core domain themes

- **Customer recovery** — identify and act on inactive / missing accounts.
- **Growth** — new acquisition, product introduction, replacement campaigns.
- **Route execution** — day-level discipline with GPS-backed visit records.
- **Targets vs achievement** — sales amount, product volume/qty, customer KPIs.
- **Field intelligence** — market research, follow-ups, outstanding collection.

04 ARCHITECTURE

4. System Architecture

CRGS follows a classic three-tier client-API-database architecture. Both the web portal and the mobile app are thin clients that call the same Flask service. The service reads and writes Oracle tables used by the wider ERP (customers, bills, items, routes) and dedicated CRGS operational tables.



Figure 1 — High-level system architecture

4.1 Deployment topology (typical LAN)

- API host example: **http://192.168.61.41:5000**
- Web Vite proxy: **/api** → **API host** (vite.config.ts)
- Mobile API base: **http://192.168.61.41:5000/api** (overridable via --dart-define=API_BASE_URL)
- Oracle DSN (config default): host/service for RFSS schema on rgc

05 TECHNOLOGY STACK

5. Technology Stack

5.1 Web Admin Portal

Technology	Version / note
React / React DOM	^19.2.7
TypeScript	~6.0.2
Vite	^8.1.0
React Router DOM	^7.18.0
Tailwind CSS (+ Vite plugin)	^4.3.1
Recharts	^3.9.0 — dashboards & reports
Framer Motion	^12.42.2
Lucide React + Radix Slot / CVA	Iconography & shadcn-style UI
Oxlint	^1.69.0

5.2 REX Mobile App (Flutter)

Technology	Version / note
Package	crgs 1.0.0+1 — “REX APP — Mobile app for sales executives”
Dart SDK	^3.12.2
flutter_riverpod	^2.6.1 — state management
go_router	^14.8.1 — navigation
dio	^5.8.0+1 — HTTP client
shadcn_ui	^0.54.0
flutter_map / latlong2 / geolocator / geocoding	Maps & GPS
fl_chart	^0.70.2 — charts
connectivity_plus / shared_preferences / lottie	Offline banner, session, visit timer

5.3 Backend API

Package	Version
Flask	3.1.0
Flask-CORS	5.0.1
oracledb	4.0.1
python-dotenv	1.0.1

06 WEB ADMIN PORTAL

6. Web Admin Portal — Modules & Features

The portal is a React SPA under **frontend/**. Routes are protected by **ProtectedRoute** + **AuthContext** (session stored in localStorage as **crgs-admin-session**). Navigation: Dashboard, User Management, Route Management, Target Management, Task Management, Reports.

6.1 Authentication

- Login with **Employee Code** + **Password** via POST **/api/auth/login**.
- Branding: “CRGS Admin / Sales Manager Portal” and full CRGS expansion.
- Portal session role set as **admin** for manager users.

6.2 Dashboard

- Live KPIs: active executives, assigned routes, monthly sales/product/customer target achievement %.
- Overdue task count.
- Charts: executive performance (bar), route performance, task-type distribution (pie).
- Recent tasks table — fed by **/api/users**, **/api/targets/***, **/api/tasks**.

6.3 User Management

- Lists sales executives from CRGS_USER where ROLECODE = 3.
- Create user (username, employee code, password).
- Activate / deactivate (FLAG A/D).
- Assign multiple routes (comma-separated ROUTE column) via modal multi-select.

6.4 Route Management

- Loads route master from TBLROUTES (**/api/routes**).
- Select executive → multi-select routes → PATCH **/api/users/routes**.
- Search by route name / number.

6.5 Target Management (3 tabs)

Tab	Table	What managers configure
Sales targets	CRGS_SALETARGET	Amount, period (daily/weekly/monthly), routes, due date
Product targets	CRGS_PRODUCTTARGET	quantity, volume, new_promotion, replacement, own_products — multi-product picker
Customer targets	CRGS_CUSTOMERTARGET	new_acquisition, missing_recovery, outstanding_collection, purchase_limit

6.6 Task Management

Campaign-style tasks assigned to executives (CRGS_TASK):

- Missing Customer Follow-up
- Outstanding Collection Follow-up
- New Product Introduction
- Product Replacement Campaign
- Customer Visit Campaign
- Own Products
- Market Research

Filter by type; create and delete via **/api/tasks**.

6.7 Reports (8 tabs)

- Executive Performance
- Route Performance
- Customer Recovery
- New Customer
- Product Sales
- Outstanding Collection
- Target vs Achievement
- Market Trend

Reports combine live target/user API data with richer analytical views (some market/recovery panels may use mock enrichment for presentation completeness).

6.8 Route map (web)

Path	Page
/login	LoginPage
/	DashboardPage
/users	UsersPage
/routes	RoutesPage
/targets	TargetsPage
/tasks	TasksPage
/reports	ReportsPage

07 REX MOBILE APP

7. REX Mobile App — Modules & Features

REX APP (package **crgs**) is the executive's day-to-day tool. Navigation uses **go_router** with a soft-rounded sidebar shell. Unauthenticated users go to login; authenticated users with incomplete onboarding see a 3-slide onboarding flow that sets **ONBOARD_FLAG** via the API.

7.1 Shell navigation

Route	Screen	Purpose
/routes	RouteListScreen	Assigned routes (default home)
/routes/:routeId	CustomerListScreen	Customers for a route
/tasks	TaskManagementScreen	Assigned campaign tasks
/orders	OrdersScreen	Expected orders list
/dashboard	DashboardScreen	Personal KPIs & today overview
/profile	ProfileScreen	Identity, routes, theme, logout
/reports	ReportsScreen	Personal charts summary

7.2 Field workflows (standalone)

Route	Screen	Purpose
/visit/:customerId	VisitTrackingScreen	GPS visit start/end + products
/customers/:id	CustomerDetailScreen	Account detail & history
/recovery/:customerId	RecoveryFormScreen	Missing-customer recovery reasons
/products/:customerId	ProductIntroScreen	Catalog / cart / expected order
/outstanding	OutstandingCollectionScreen	Invoice collection workflow
/follow-up	FollowUpScreen	Calendar + timeline follow-ups
/market-research	MarketResearchScreen	Field intelligence capture
/new-customer	NewCustomerScreen	New lead / prospect form
/settings	SettingsScreen	Sync, GPS, notifications, offline

7.3 Customer triage

- Tabs: **All / Missing / Outstanding** with search debounce and infinite scroll.
- Missing windows: 7 / 15 / 30 / 60 days / All (not billed today).
- Priority color coding: missing (red), outstanding (orange), follow-up (teal), regular (green).
- Detail sheets show purchase age, credit, last bills / line items from Oracle.

7.4 Visit tracking (core field feature)

- Auto start visit → POST `/api/visits/start` (CRGS_VISITDETAILS).
- GPS via LocationService + interactive map picker (flutter_map).
- Lottie-powered visit timer for presence/duration.
- Visit reason, remarks, follow-up date; ordered products section.

- End visit → POST /api/visits/end with duration and location.

7.5 Orders & product introduction

- List expected orders from CRGS_ORDERHDR / ORDERDTL.
- Product intro: recommended / alternative products, cart, expected-order capture.
- Item master sourced from /api/items (ITEMMASTER).

7.6 Dashboard & supporting modules

- Hero: monthly %, route performance, pending tasks; today's target overview.
- Today's route map, weekly chart, Up Next, quick actions; FAB for New Lead.
- Tasks: All / Route / Additional / Follow-up progress with status badges.
- Outstanding, recovery, market research, follow-ups, settings (theme, connectivity banner).

08 BACKEND & DATA

8. Backend API & Data Layer

Service id: **crgs-admin-api**. Factory in backend/app/__init__.py registers blueprints, CORS, Oracle pool, and health endpoints GET /api/health and GET /api/health/db.

8.1 API blueprints

Prefix	Module	Key endpoints
/api/auth	auth.py	POST /login, PATCH /onboarding
/api/users	users.py	GET /, POST /, PATCH /routes, PATCH /status
/api/routes	routes_data.py	GET / (search, pagination)
/api/targets	targets.py	/sales, /products, /customers — GET/POST/DELETE
/api/items	items.py	GET / ITEMMASTER
/api/tasks	tasks.py	GET /, POST /, DELETE /
/api/visits	visits.py	POST /start, POST /end
/api/orders	orders.py	GET /, POST /
/api/customers	customers.py	list, stats, last-purchase, last-order, bill-items

8.2 Oracle tables & views

Entity	Default object	Use
Users / auth	CRGS_USER	Login, roles, routes, onboard flag
Routes	TBLROUTES	Route master
Customers	CUSTOMERS + CUSTOMERAGEVIEW	List, missing age
Billing	BILLHDR / BILLDTL	Last purchase, line items
Items	ITEMMASTER	Product catalog
Sales targets	CRGS_SALETARGET	Sales KPIs
Product targets	CRGS_PRODUCTTARGET	Product KPIs
Customer targets	CRGS_CUSTOMERTARGET	Customer KPIs
Tasks	CRGS_TASK	Field campaigns
Visits	CRGS_VISITDETAILS	Visit start/end
Orders	CRGS_ORDERHDR / ORDERDTL	Expected orders

Missing-customer threshold defaults to **MISSING_DAYS = 30** (configurable). Customer API supports priority filters, pagination, and route-level stats.

09 DOMAIN & ROLES

9. Domain Model & User Roles

9.1 Key entities

- **User / Executive** — employee code, routes, FLAG, ONBOARD_FLAG, roleCode.
- **Route** — code, area, assigned executive, customer count.
- **Customer** — GPS, credit, purchase age, priority (missing / outstanding / follow-up / regular).
- **Targets** — SalesTarget, ProductTarget, CustomerTarget with period & achievement.
- **Task** — type, status, priority, assignment to employee/route.
- **Visit** — start/end timestamps, location, reason, ordered products.
- **Order** — header + line items (expected order capture).
- **Follow-up / Outstanding / Market Intel / Lead** — supporting field objects.

9.2 User roles

Role	Where	Capabilities
Admin / Sales Manager	Web portal (session role admin)	Manage users, routes, targets, tasks, dashboards, reports
Sales Executive	Oracle ROLECODE = 3; REX APP	Execute assigned routes, visits, orders, tasks; view personal targets

- FLAG = **A** active, **D** inactive.
- ONBOARD_FLAG = **Y** when mobile onboarding is complete.

10 OPERATIONAL FLOW

10. End-to-End Operational Flow

For presentation audiences, the platform is best explained as a closed loop:

Step	Action
1. Provision	Manager creates executive account and activates FLAG = A (Web → Users).
2. Assign territory	Manager assigns one or more routes from TBLROUTES (Web → Routes).
3. Set goals	Manager creates sales, product, and customer targets for the period (Web → Targets).
4. Launch campaigns	Manager assigns tasks (missing follow-up, product intro, etc.) (Web → Tasks).
5. Field login	Executive logs into REX, completes onboarding once, sees assigned routes.
6. Triage day	Executive opens a route, filters Missing / Outstanding, prioritizes stops.
7. Execute visit	Start GPS visit, capture reason/remarks/products, end visit with duration.
8. Capture demand	Introduce products and save expected orders tied to the customer/visit.
9. Measure	Manager dashboard & reports reflect achievement; executive sees personal KPIs.

This loop is the primary story for demos: **plan on the web, execute on mobile, measure from shared Oracle data.**

11 UX / UI

11. UX / UI Design Principles

11.1 Web portal

- Dark teal sidebar with light content canvas (gray-50).
- Primary accent ~ #00766e / #0F766E aligned with mobile brand.
- Stat cards, Recharts visualizations, status badges, modal forms.
- Responsive sidebar drawer on narrow viewports.
- Multi-select pickers for routes and item master products.

11.2 REX mobile

- Teal brand (#0F766E) + orange accent for CTAs / login hero.
- shadcn_ui + Material hybrid (ShadApp.router).
- Soft rounded sidebar (28px) with staggered entry animations.
- Fade page transitions; Lottie visit timer; interactive maps.
- Priority color coding for customer cards; offline connectivity banner.
- Light / dark theme toggle; infinite-scroll lists; pull-to-refresh dashboard.

12 INTEGRATION

12. Integration Map (Web ↔ Mobile ↔ Oracle)

Capability	Web	Mobile	Shared API / table
Login	Yes	Yes	POST /api/auth/login → CRGS_USER
Onboarding	—	Yes	PATCH /api/auth/onboarding
Create / activate executives	Yes	—	/api/users
Assign routes	Yes	Reads assigned	CRGS_USER.ROUTE + TBLROUTES
Set / read targets	Set	Read (dashboard)	/api/targets/*
Assign / list tasks	Assign	List / execute	/api/tasks → CRGS_TASK
Customers / missing	Reports	Core field flow	/api/customers* + age views
Visits	Monitor via reports	Start / end	/api/visits → CRGS_VISITDETAILS
Orders	—	Capture & list	/api/orders → ORDERHDR/DTL
Product master	Target picker	Intro / orders	/api/items → ITEMMASTER

13 MATURITY & ROADMAP

13. Current Maturity & Roadmap Notes

13.1 Production-ready core (API-backed)

- Authentication, users, route assignment, targets, tasks.
- Customers (list, stats, last purchase/order, bill items).
- Visits start/end, expected orders, item master.
- Manager dashboard KPIs and executive mobile route/customer/visit flows.

13.2 UI-first areas (presentation-ready; deeper persistence next)

- Some mobile screens (market research submit, recovery form save, new-lead persistence, parts of outstanding/follow-up) are UI-complete and ready for full API wiring.
- Certain web report tabs may blend live targets with mock enrichment for market/risk storytelling.

13.3 Suggested next increments

- Persist recovery outcomes, market research, and leads to Oracle.
- Push notification / reminder for overdue tasks and commitment dates.
- Hardened JWT/session auth beyond employee-code login if required by IT policy.
- Offline queue for visit/order capture when connectivity drops.
- Expand reports with fully live customer-recovery and collection analytics.

14 DEMO SCRIPT

14. Demo Script for Presentation

Use this sequence for a ~10–12 minute live walkthrough:

#	Surface	Talk track
1	Slide / title	Introduce CRGS vs REX audiences and the recovery + growth mission.
2	Web Login	Show manager portal branding and employee-code login.
3	Dashboard	Point to live KPIs: executives, routes, target %, overdue tasks.
4	Users + Routes	Create/select executive; assign routes; explain ROLECODE 3.
5	Targets + Tasks	Create one sales target and one missing-customer task.
6	REX Login	Log in as that executive; complete/skip onboarding if already done.
7	Routes → Customers	Open route; switch Missing tab; open a customer.
8	Visit	Start visit, show GPS/map/timer, add remarks, end visit.
9	Products / Orders	Introduce a product and capture an expected order.
10	Web Reports	Return to portal; show Target vs Achievement / Executive Performance.
11	Close	Restate plan → execute → measure loop and roadmap items.

15 CONCLUSION

15. Conclusion

CRGS delivers a practical, ERP-connected platform for sales organizations that need both **office control** and **field execution**. The web portal equips managers to staff routes, set multi-dimensional targets, and launch recovery/growth campaigns. REX APP equips executives to prioritize the right customers, prove visits with GPS, and capture demand in the moment.

Because both clients share one Flask API and one Oracle schema, every assignment on the portal has a clear field counterpart — and every field action can feed manager visibility. That shared operational loop is the core value of the Customer Recovery & Growth System.

Deliverables covered in this report

- Web Admin Portal feature catalog (6 modules + reports)
- REX Mobile App feature catalog (shell + field workflows)
- Architecture, tech stack versions, API & Oracle map
- Roles, domain model, integration matrix, demo script

Document generated 13 July 2026 • CRGS-Admin repository